

Every edit stops the playtest

THE USER

Technical designers at small studios iterate on playable prototypes with real playtesters in the room.

THE COST

Every stop costs the thing they are trying to observe: how people actually play.

Beetle's promise

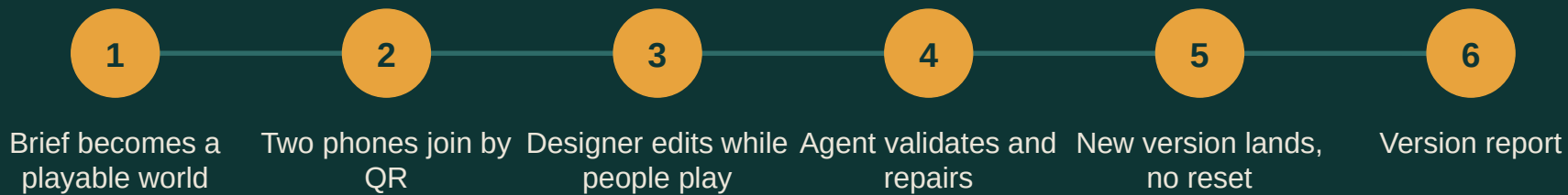
Change the game without stopping the game.

TODAY'S LOOP



Each turn of the loop loses the live playtest you were trying to watch.

The Beetle workflow



WHAT IT RUNS ON

One machine

a single GB10 on the studio LAN

One local model

Ollama on loopback

One real agent runtime

OpenClaw with Beetle tools

Zero cloud

no fallback exists

The working video



[VIDEO PLACEHOLDER]

Insert the rehearsal recording here (90 to 120 s). Not yet recorded at 13:40 CDT.

The caption names the agent mode used; any time compression is labelled on screen

ON-SCREEN CAPTIONS

0 to 6 s Beetle. Change the game without stopping the game.

6 to 22 s A local model composes the world. Validated by game code before it is published.

22 to 34 s Two players join from their phones. Same screen, their phones only send controls.

34 to 58 s Edits land live. No disconnect, no reset.

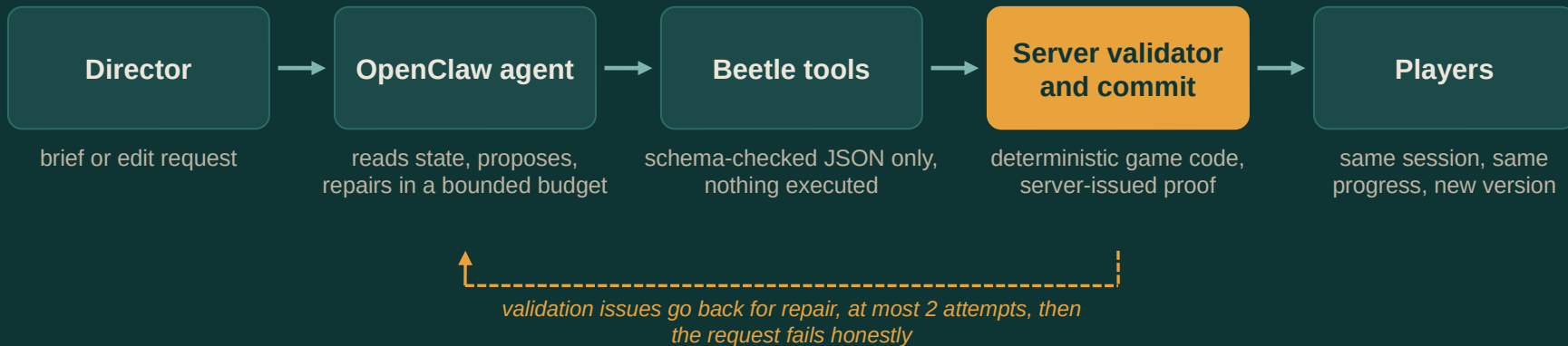
58 to 90 s The validator refuses a world nobody can finish (DISCONNECTED_GOAL). The agent repairs it and commits only a passing version.

90 to 105 s Measured on one GB10. Local model, local agent in [direct or OpenClaw] mode, no cloud.

105 to 115 s Beetle

Fresh world, two phones, a live edit, preserved progress, a conflicting edit caught by the validator, the repaired version committed.

Agent architecture and validation



TWO KINDS OF STATE

- WorldSpec is structure. It changes only when a transaction commits, one version per commit.
- SessionState is players, relics, score. The simulation advances it; the agent never writes it.
- Commit needs a server-issued proof.

VALIDATION IS GAME CODE

- Geometry, bridge sockets, walk field, reachability with the gate locked, occupied support at commit time.
- An LLM saying valid is never approval.
- Nothing from the model runs. Output is schema-checked JSON only.

Measured local-first results



MODEL

qwen3.5:4b, Q4_K_M

4.7B, 3.4 GB, Ollama 0.35.1 on loopback. qwen3.8:27b not benchmarked: pull failed once, re-downloading. docs/MODEL_SELECTION.md

BRIEF, WARM, QUIET GPU

26.2 / 26.5 / 27.7 s

min / p50 / max, 3 runs, first draft valid (bench-qwen3.5_4b-1791049348895.json). 10 s target not met. 14 s once after normalization (RESULTS.md run 3).

COLD / CONTENTED GPU

23.1 s load; 58 to 214 s

cold: 23.1 s model load in a 54.4 s first draft (probe-run.log). Contended: drafts 58.8 to 214.1 s, one edit 45.1 s (bench-qwen3.5_4b-1791050596040.json).

EDIT TO COMMIT

2.0 / 8.0 / 14.0 s

min / p50 / max, 6 edits, 5 of 6 committed; run 3 after normalization, quiet GPU, direct mode (run-1791052136845.json). Run 2 before: 2 of 6, 3.0 s each.

INVALID EDITS CAUGHT

6 of 12 refused

runs 2 and 3 (RESULTS.md); repairs that then committed: 1 (run 3 E6); invalid worlds committed: 0. On garden5 a DISCONNECTED_GOAL refusal was repaired and committed in 3.0 s.

NORMALIZATION EFFECT

2 of 30 to 13 of 30

corpus validity without vs with the deterministic normalizer (MODEL_FAILURE_MODES.md, corpus/summary.json). Live edits committed: 2 of 6 to 5 of 6.

LATENCY, LOOPBACK

0.48 ms RTT, 18.8 ms tick

p50, scripted controllers (p95 0.81 and 33.1 ms; LATENCY.md). Not input to photon: no phone, Wi-Fi, browser or display included.

SESSION CONTINUITY

not measured live

no player connected in runs 2 and 3. Preserved ids, relics, score and reconnect are covered by integration tests with real sockets (BUILD_STATUS.md 10, 12).

OFFLINE PROOF

not run

only a before record exists (data/offline-proof/1791049273707.json: egress not blocked, server not up). Procedure: docs/OFFLINE_PROOF.md. Nothing claimed.

All timings are from the direct harness, labelled [direct] in every report, and each names its GPU state because the Ollama daemon was shared for most of the day. OpenClaw tool calls are proven against a fake server (packages/agent/SMOKE.md); the run against the real server times out before any model call and is being diagnosed.

Business hypothesis and next validation



HYPOTHESIS

- Studios will pay for faster prototype iteration that keeps playtests running.
- Their data stays on their machines, which matters to studios with unreleased IP.

NOT CLAIMED

- Market size.
- Pricing.
- Customer traction.
- Speed-up percentages without a measured baseline.

NEXT VALIDATION

- Time a conventional edit-rebuild-regather loop with the same designers and the same change list.
- Compare it against Beetle's measured edit-to-commit times.



Change the game without stopping the game.